Semantic Gravitational Attractors

Canonical Identity, Regime-Relative Convergence, and the Stabilization of Meaning

Semantic gravitational attractors can reduce duplication, identity churn, and unsupported divergence.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Content-addressed systems make an important move: they replace positional identity with identity derived from a representation. When that representation is first normalized under declared rules, many variant inputs can converge upon one canonical form. This paper develops **semantic gravitational attractors** as a bounded architectural metaphor for the systems that deliberately create and govern that convergence.

The attractor is not a physical force, a universal semantic equivalence oracle, or proof that meaning can be reduced to bytes. It is a policy-and-infrastructure arrangement: a versioned canonicalizer, a declared equivalence regime, domain-separated hashing, governed admission, lineage, constrained transformation, and evidence together route admitted representations toward stable identities. The convergence is only as sound as the distinctions preserved by the regime.

The paper therefore replaces the strong claim that "equivalent meaning collapses" with a testable, regime-relative claim. It distinguishes false merges from false splits, identifies cases for which canonicalization is undefined or refused, separates byte identity from reference and standing, and states the environmental closure required for deterministic replay. It also integrates canonical identity with MCIR, GMB, the universal GSM, governed project lineage, semantic topology, binding, debt, receipts, and semantic computing infrastructure.

Semantic gravitational attractors can reduce duplication, identity churn, and unsupported divergence. They can also make a mistaken equivalence more durable, erase a material distinction, or centralize an illegitimate canonicalization law. The resulting architecture is valuable precisely when it keeps those risks inspectable and revisable.

Source, Evidence, and Standing

This is an AI-assisted integrated edition derived from one exact author manuscript. Its custody manifest is
`work/paper-saturation-v1/evidence/semantic-gravitational-attractors/integration-custody-manifest.v0.1.json`,
SHA-256
`B85D67F55D95D310191FD3CBD1A9BB44F61A4EE28DC18A3B1A38B86666F4274B`.
The source remains unchanged.

The attractor concept and its principal architectural applications are author material. Formal limits, corpus interfaces, tests, counterevidence, and publication decisions added here are AI editorial or synthetic candidates until adopted by the author. No external research was conducted. The source's examples of Git, blockchain, Nix, and IPFS remain unverified illustrations.

1. The Bounded Thesis

Conventional systems often bind identity to a path, database key, object address, or mutable record. Copies and transformations can then persist without an explicit account of whether they are equivalent, successors, forks, or unrelated objects. This produces not merely duplication but an identity-governance problem.

A semantic gravitational attractor is an arrangement under which admitted representations are normalized, assigned content-derived identity, connected to their provenance and transformation history, and subjected to one declared admission and revision regime. It creates convergence pressure because multiple admitted inputs may be routed to one representative.

The bounded thesis is:

> Versioned canonicalization plus governed identity and lineage can stabilize a declared class of semantic representations under a declared equivalence regime.

This thesis does not establish identity of meaning in every context. It establishes an inspectable decision about which distinctions a system treats as material.

2. Canonicalization Is a Regime

Let X be a representation space and let

$$C_v : X \rightharpoonup B$$

be a partial canonicalizer under versioned regime v. The partial arrow matters. Some inputs may be outside the domain, malformed, ambiguous, contested, or

unsafe to normalize.

The regime induces an equivalence relation only on admitted inputs:

$$x \sim_v y \quad \Longleftrightarrow \quad C_v(x) = C_v(y).$$

An identity may then be constructed with domain separation:

$$ID_v(x) = H(\texttt{domain} \parallel v \parallel C_v(x)).$$

Version and domain must be bound to the identity. Otherwise the same bytes produced under different semantic laws may be treated as interchangeable. Cryptographic collision resistance is an assumption of the identity mechanism, not a proof of semantic equivalence.

The basin associated with canonical value b is

$$\mathcal{B}_v(b) = \{x \in \mathrm{Dom}(C_v) : C_v(x) = b\}.$$

This basin is the precise architectural object behind the attractor metaphor. The system causes admitted representations in the basin to share one identity. It does not show that all observers, purposes, or jurisdictions should consider them equivalent.

3. Four Identity Layers

At least four identity layers must remain distinct:

1. **representation identity:** exact bytes or syntax of the supplied artifact;
2. **canonical identity:** bytes after applying C_v;
3. **regime-relative semantic identity:** the equivalence asserted by $\sim_v$;
4. **governed standing:** the authority, evidence, jurisdiction, and status under which the identity may be used.

Two representations can share a canonical identity yet differ in custody or authority. Two records can refer to the same external object while remaining different claims. A canonical identifier can be stable while the interpretation, resolver, registry, or institution around it changes.

The architecture therefore needs more than a digest. It needs the regime version, source custody, declared semantics, authority, evidence, conflict state, and revision lineage.

4. False Merges, False Splits, and Refusal

A canonicalizer can fail in two opposite directions.

A **false merge** occurs when $C_v(x) = C_v(y)$ even though a distinction between x and y is material for the declared purpose. Removing authorship, order, units, jurisdiction, negation, or uncertainty can cause this failure.

A **false split** occurs when $C_v(x) \neq C_v(y)$ even though the regime intends the inputs to be equivalent. Insignificant whitespace, field order, aliases, serialization choices, or equivalent expressions may cause this failure.

A third category is essential: **undefined or refused canonicalization**. A system should not force a result when an identity is contested, an input exceeds the regime, or the relevant distinction cannot yet be represented. The output space is therefore not simply old identity or new identity. It includes admit, branch, defer, refuse, and escalate.

These error classes make the attractor evaluable. They also show why stronger convergence is not automatically better. An attractor that hides false merges can produce coherent-looking semantic damage.

5. Referential Stability Is Conditional

Content-derived identity can make a reference independent of storage location. The digest can survive file movement and duplication. That is valuable but narrower than permanent reference.

Reference stability depends on:

- the continued availability of the identified bytes;
- the preservation of the hash and canonicalization algorithms;
- the version and domain interpretation of the identifier;
- the integrity of resolvers and registries;
- the persistence of external referents where claims concern them; and
- the survival of authority, evidence, and standing bindings.

An immutable identifier can persist after its interpretation becomes obsolete or its authority is revoked. The claim should therefore be that canonical identity stabilizes a representation reference under stated dependencies, not that all reference remains stable across time and transformation.

6. Deterministic Replay Requires Closure

The source proposes $State = Replay(EventLog)$. A more complete relation is:

$$S_t = R(E_{0:t}, I, D, K, P, T, N),$$

where E is the event sequence, I the initial state, D dependencies, K implementation and interpreter versions, P configuration and policy, T time and ordering inputs, and N nondeterministic or external inputs.

Replay is deterministic only to the extent that these dependencies are captured, eliminated, simulated, or bounded by an equivalence definition. Content-addressed artifacts help reconstruct inputs, but they do not close clocks, networks, randomness, concurrency, external services, hardware behavior, or human decisions by themselves.

Replay evidence should report exact reproduction, admitted equivalence, divergence, missing dependency, or refusal. A hash match for some artifacts does not by itself prove state equivalence.

7. Governed Transformation and the SIS Boundary

The source proposes a Semantic Instruction Set (SIS) so transformations take the form:

$$Instruction \rightarrow Event \rightarrow CanonicalArtifact.$$

This can reduce uncontrolled mutation if every material effect path passes through the governed interface. The stronger invariant is:

> No admitted identity or effect may be created or changed outside a path whose authority, inputs, canonicalization regime, obligations, result, and evidence are inspectable.

That invariant remains unproved until bypass paths are tested: direct storage writes, alternate APIs, imports, repair tools, administrator actions, migrations, caches, failed transactions, and cross-system translations. Constrained mutation makes drift detectable and resistible only within its demonstrated coverage.

8. Versioning, Migration, and Successor Identity

Changing canonicalization law changes the induced equivalence relation. A new version can split an old basin, merge several basins, reject previous values, or expose a distinction that the old regime erased.

A migration therefore needs:

- old and new regime identifiers;
- a mapping or explicit absence of mapping;
- split and merge records;
- preserved original custody;
- transformed outputs and remainders;
- the authority for adopting the new law;
- impact analysis for references and dependent artifacts; and

- receipts for decisions and verification.

Historical identity should not be rewritten. The new regime creates a successor domain connected by typed lineage. This turns canonicalization change from silent identity drift into a governed transition.

9. The Limits of Gravity, Conservation, and Event Horizons

"Gravity," "mass," "field," "equilibrium," "conservation," and "event horizon" are useful images only when their architectural referents are named.

- attraction means routing admitted variants to a representative;
- mass may describe accumulated references or dependencies, but no conserved quantity is defined;
- a field may describe a graph of identities and transformations, not a physical field;
- equilibrium may describe declining identity churn under fixed rules, not thermodynamic equilibrium;
- an event horizon may describe an admission boundary, but crossing it is neither irreversible nor physically causal; and
- conservation may describe preservation of declared obligations or lineage, not conservation of meaning.

The source claim that the attractor is "not metaphorical" should be revised. The mechanism is architectural; gravity is the metaphor used to reason about its convergence behavior. No semantic conservation law is established in this paper.

10. GMB, GSM, MCIR, and the Corpus

The **Governed Meaning Base (GMB)**, current in the `CCF -> CMB -> GMB` lineage, can retain semantic identities together with their regimes, provenance, standing, authority, evidence, conflicts, gaps, and revision history. Inclusion in GMB does not validate an equivalence or grant standing.

The universal **Governed Semantic Model (GSM)** supplies typed coordinates for concepts, claims, systems, methods, and relations. Semantic Manifold Topology and Governed Semantic Cartography help describe neighborhoods, dimensions, and paths without turning geometric proximity into identity.

MCIR can carry a canonical, machine-legible representation through governed transformation. It must preserve the constraints and unresolved remainders that the attractor regime declares material. **Meaning as Binding** clarifies that identity becomes consequential through admitted bindings. **Semantic Debt** identifies the accumulated cost of unresolved aliases, conflicts, untyped relations, and migration obligations. **Reasoning by Way of Receipts** supplies the evidence pattern for recording proposals, decisions, effects, and verification. **Governed Semantic Project Lineage** extends content addressability into custody and successor relations. **Semantic Computing Infrastructure** situates

these mechanisms in a wider computational layer.

These are directional interfaces, not identity claims or automatic mergers.

11. Realization Funnel

$$\begin{aligned} & P \\ & \to S \\ & \to CM_1 \\ & \to CM_2 \\ & \to IC \\ & \to IR \\ & \to E \\ & \to R \end{aligned}.$$

- **Possibility:** enumerate canonicalization regimes, equivalence policies, identity layers, and admission boundaries.
- **Specification:** define C_v, $\sim_v$, domain-separated identity, invariants, errors, lineage, and required evidence.
- **CM1:** retain technically admissible regimes, including undesirable or unsafe ones, with their failure surfaces visible.
- **CM2:** prefer regimes that minimize consequential false merges and false splits while preserving custody, authority, accessibility, revision, and refusal.
- **IC:** select an **Instance Constitution** or **Instance Configuration** for a bounded domain. The expansions are interchangeable.
- **IR:** produce an **Instance Realization** or **Implementation Realization** such as a canonicalizer, registry, resolver, GMB profile, migration tool, or verifier. The expansions are interchangeable.
- **Evidence:** measure conformance, coverage, error rates, replay divergence, identity churn, reference survival, obligation carriage, and authority correctness.
- **Revision:** amend or replace the regime through explicit successor identity and migration receipts.

This paper reaches Specification and proposes CM1/CM2 selection criteria. It does not select a complete IC, produce an IR, or establish conformance evidence.

12. Evaluation Program

An implementation should report at least:

1. false-merge rate on distinctions declared material;
2. false-split rate on equivalences declared intended;
3. canonicalization coverage, refusal, and escalation rates;
4. identity churn across versions and migrations;
5. replay divergence under a declared environment closure;

6. reference survival and resolver correctness;
7. provenance and custody completeness;
8. obligation carriage through transformations;
9. authority and standing correctness; and
10. human ability to understand, challenge, and revise the equivalence law.

No one measure is "semantic gravity." Each supports a bounded property claim.

Convergence-Regime Role

The completed P1 corpus supports a narrower and more useful placement for Semantic Gravitational Attractors (SGA): it is a **convergence-regime profile** governing when representations may be normalized toward a shared semantic identity. It is not a new meaning substrate, identity primitive, storage layer, or general theory of binding.

- **Adjacent work:** *Constitutional Binding Theory*
  Supplies: General record of manifold, anchor, authority, exclusions, evidence, and revision.
  SGA boundary: SGA is one mechanism for governed many-to-one binding under an equivalence regime.
- **Adjacent work:** *Meaning as Binding*
  Supplies: Relational account of meaning and consequential attachment.
  SGA boundary: SGA stabilizes selected representations; it does not reduce all meaning to canonicalization.
- **Adjacent work:** *Identity Spine Doctrine*
  Supplies: Admission, split, merge, version, and correction discipline.
  SGA boundary: SGA proposes convergence pressure; Identity Spine governs whether convergence may become corpus identity.
- **Adjacent work:** *Governed Semantic Project Lineage*
  Supplies: Version families, custody, successor and fork relations.
  SGA boundary: SGA depends on lineage when canonicalization rules or identities change.
- **Adjacent work:** GMB and MCIR
  Supplies: Custody/retrieval substrate and canonical machine representation.
  SGA boundary: Neither storage nor serialization proves that the selected equivalence is semantically adequate.

The reusable convergence object is:

$$G = (J, C_v, \sim_v, A_v, L_v, X_v),$$

where J is the jurisdiction, C_v the versioned canonicalizer, $\sim_v$ the declared equivalence relation, A_v the admission and authority policy, L_v lineage and

migration rules, and X_v the distinctions and semantic remainder intentionally preserved outside convergence. A digest of $C_v(x)$ identifies the canonicalized result only under this envelope.

This makes the failure boundary explicit. A **false merge** loses a material distinction; a **false split** prevents legitimate convergence; a **false refusal** rejects an admissible representation; and a **silent remainder loss** converges bytes while dropping an obligation, disagreement, uncertainty, or provenance relation. The fourth class is added by corpus integration because MCIR, constraint carriage, and GMB all require more than equivalence of normalized form.

The universal GSM should type `CANONICALIZES_UNDER`, `EQUIVALENT_UNDER`, `PRESERVES_DISTINCTION`, `LOSES_REMAINDER`, `SUCCESSOR_REGIME_OF`, `MIGRATES_IDENTITY_TO`, and `REFUSED_BY`. No unqualified `SAME_MEANING_AS` edge should be inferred from a shared attractor.

This integration establishes four editorial commitments: adopt convergence-regime profile as SGA's official corpus role; adopt the six-part regime object; add silent remainder loss to the primary failure taxonomy; and decide whether the gravitational metaphor remains in the title once the mechanism is presented directly.

Boundary-Safe, Perspective-Preserving, Counteroriented, and Authority-Aware Convergence

The fixed-seed comparison with The Path Was Open, The Locus of Shared Experience, Recursive Semantic Orientation, and Split-Authority AI Coordination turns a convergence regime into a visibly governed intervention:

$$\begin{aligned} G^* = (&jurisdiction, canonicalizer, equivalence, admission,\\ &lineage, preserved_distinctions, boundary_classes,\\ &perspective_remainder, orientation_checks,\\ &authority_topology, contest, revision) \end{aligned}.$$

The Path Was Open shows that normalizing several concrete effects into one abstract class can hide a bypass, actor difference, or feasible precaution. A convergence regime must declare what distinctions remain accountability-relevant. The Locus of Shared Experience adds participant-indexed meaning, permission, correction, and unresolved phenomenal difference that cannot be optimized away as disagreement noise.

Recursive Semantic Orientation treats attractors as possible investigation biases as well as useful compression. An investigator should retain countermodels, excluded regions, unknowns, and probes that can reveal a false merge or false refusal. Convergence of orientation is not optimality or truth.

Split-Authority AI Coordination supplies an authority boundary: mandate, specification, realization, verification, and effect roles may need shared identifiers without sharing binding authority. Canonical artifacts and common schemas can create a common-mode failure or authority-laundering path if role-specific standing and negative power are collapsed.

The universal GSM can type
`CANONICALIZES_EFFECT_CLASS_WITH_REMAINDER`,
`PRESERVES_PARTICIPANT_PERSPECTIVE`, `COUNTERORIENTED_BY_PROBE`,
`SHARES_IDENTIFIER_NOT_AUTHORITY`, `CREATES_COMMON_MODE_CONVERGENCE`,
`CONTESTS_EQUIVALENCE_REGIME`, and
`REVISES_AFTER_FALSE_MERGE`.

This randomized review establishes four editorial commitments: expand the regime object with boundary, perspective, orientation, and authority dimensions; require accountability and participant remainder; add counterorientation tests; and prevent canonical convergence from collapsing role authority or verification independence.

Counterorientation and Perspective-Preserving Convergence

Six downstream uses divide semantic convergence into beneficial stabilization and common-mode risk. Semantic Binding Quality and transformation-dominant diagnostics evaluate what a convergence regime preserves, changes, or erases. Recursive Semantic Orientation requires countermodels against false convergence. Split-Authority AI Coordination treats shared canonicalization as a possible common-mode defect. *The Locus of Shared Experience* requires participant-indexed difference to survive convergence. *The Path Was Open* shows that false merging of actors, effects, or notices can erase an accountability distinction.

The attractor profile should therefore expand to:

$$\begin{aligned} A^* = (&subject, inputs, equivalence_regime, canonicalizer,\\ &authority, admission, observers, perspectives, preserved,\\ &transformed, erased, remainder, countermodels,\\ &anti_attractors, bypass, migration, evidence, contest,\\ &successor) \end{aligned} .$$

An `anti_attractor` is not noise added for balance. It is a protected counterexample, dissenting representation, minority case, alternate ontology, or deliberately noncanonical view retained to test whether convergence is destroying a material distinction. Counterorientation asks what evidence would pull the system away from its present canonical basin. A regime unable to preserve or revisit such cases risks becoming self-sealing.

Perspective preservation is property-specific. Several representations may lawfully converge for retrieval while remaining distinct for authority, responsibility, experience, consent, or evidentiary standing. Shared output, terminology, or hash identity cannot establish agreement, truth, independent verification, or a shared subject.

The universal GSM can type
`CONVERGES_UNDER_EQUIVALENCE_REGIME`,
`RETAINS_ANTI_ATTRACTOR`, `PRESERVES_PARTICIPANT_DIFFERENCE`,
`CREATES_COMMON_MODE_SEMANTIC_RISK`, `COUNTERORIENTS_FROM_CANONICAL_BASIN`,
`FALSE_MERGES_ACCOUNTABILITY_SUBJECT`, and
`REOPENS_CONVERGENCE_REGIME`.

This reverse-dependency review establishes four editorial commitments: add anti-attractors and counterorientation to the architecture; require perspective- and property-specific preservation; treat convergence as a possible common-mode dependency; and test accountability, authority, and experiential distinctions against false merge.

13. Counterevidence and Failure Modes

The proposal should be narrowed or rejected where: (1) semantic equivalence is assumed universal; (2) canonicalization is total only by silently dropping unresolved inputs; (3) a false merge erases authorship, jurisdiction, units, negation, uncertainty, or evidence; (4) a false split defeats intended interoperability; (5) hash collision assumptions are unstated; (6) domain separation is absent; (7) regime version is not identity-bound; (8) canonical bytes are treated as truth; (9) identity is confused with external reference; (10) GMB inclusion is treated as standing; (11) a registry becomes a centralized oracle without contest; (12) historical identifiers are rewritten during migration; (13) split and merge lineage is lost; (14) replay omits code, dependencies, configuration, order, time, randomness, or external inputs; (15) exact-byte replay is confused with behavioral equivalence; (16) the SIS has bypass paths; (17) failed or partial transformations leave unreceipted state; (18) a resolver returns the wrong domain or version; (19) unavailable content makes a stable identifier unusable; (20) transformations drop obligations; (21) representations converge while interpretations diverge; (22) an obsolete identity remains authoritative by inertia; (23) the gravity metaphor is mistaken for physics; (24) conservation language implies an undefined quantity; (25) event-horizon language implies irreversibility; (26) the technology examples fail external verification; (27) the regime creates excessive governance or accessibility cost; (28) domain participants reject the encoded equivalence; (29) adversarial inputs exploit normalization ambiguity; (30) a system cannot express relevant exceptions; or (31) the architecture merely relocates semantic judgment while claiming to eliminate it.

Stabilize Meaning Without Building a Semantic Singularity

Public orientation

Semantic Gravitational Attractors describes how a declared canonicalization and governance regime can route variant representations toward stable identities. This can reduce duplication and identity churn. It can also stabilize a mistake, erase a minority distinction, or make one institution's equivalence law look universal. The attractor is therefore a governed convergence mechanism, not a discovery engine for meaning itself.

Adversarial tests

1. **False-merge attack.** Authorship, negation, units, jurisdiction, uncertainty, participant identity, or evidentiary standing is erased while inputs converge.
2. **False-split attack.** Incidental serialization or lexical difference prevents an equivalence the declared purpose requires.
3. **Canonicalizer-capture attack.** Control of the normalization law becomes control over what can be recognized, retrieved, or admitted.
4. **Anti-attractor suppression attack.** Counterexamples, dissent, alternate ontologies, and noncanonical perspectives are treated as noise and disappear from evaluation.
5. **Migration attack.** A new regime silently rewrites prior identities instead of creating explicit successor, split, merge, or refusal lineage.
6. **Replay-closure attack.** Stable artifact hashes conceal missing clocks, dependencies, randomness, services, interpreters, or human decisions.
7. **Standing-inertia attack.** A stable canonical identity remains authoritative after its evidence, law, interpretation, or authorization has expired.
8. **Metaphor-reification attack.** Gravity, conservation, basin, or event-horizon language is treated as a physical mechanism or proof of irreversibility.

Claim and evidence ceiling

The paper specifies regime-relative convergence, partial canonicalization, layered identity, migration, refusal, and counterorientation. It does not establish universal semantic equivalence, truth, standing, permanent reference, deterministic replay, physical law, or desirable convergence. Hash identity and canonicalization prove only the distinctions encoded by the declared mechanism.

Claims about Git, blockchain systems, Nix, IPFS, cryptographic security, canonicalization standards, retrieval improvement, interoperability, scale, or user comprehension require current external sources and exact evaluations. No external research was added in this pass.

Reader-facing consequence

The useful public question is not merely how to make many representations converge. It is how to make the equivalence law visible, preserve the distinctions it could destroy, retain counterexamples, and reopen the convergence regime when it proves wrong.

15. Conclusion

Semantic gravitational attractors name a real and useful architectural pattern: admitted variants are routed through a versioned canonicalization and governance regime toward persistent identities. The mechanism can reduce duplication, stabilize transformation graphs, preserve lineage, and make semantic change more inspectable.

Its power is conditional. A canonicalizer does not discover meaning in the abstract; it enacts an equivalence law for a bounded purpose. The attractor is therefore only as trustworthy as its preserved distinctions, admission boundary, authority, evidence, bypass coverage, migration discipline, and capacity for challenge.

The deepest contribution is not that meaning becomes literally gravitational. It is that convergence itself becomes specifiable, testable, receipted, and revisable. Meaning can then be stabilized without pretending that identity, truth, authority, reference, and realization are the same thing.

Reference and Verification Note

No external source was searched in this pass. For a future evidence-qualified revision, an authorized review should verify the source-supplied descriptions of Git, blockchain systems, Nix, and IPFS; the cryptographic assumptions attached to hash identity and domain separation; relevant canonicalization standards; and any claim that a concrete system provides semantic rather than byte-level equivalence.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Cm1.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Evidence.** This stage is outside the present paper's asserted contribution and evidentiary scope.

- **Revision.** The present paper does not claim a complete revision treatment. That stage remains successor work and cannot be inferred from the architecture described here.

Corpus-Wide Reference Positioning

This paper argues that content-addressed systems make an important move: they replace positional identity with identity derived from a representation. Its nearest external comparison fields are semantic, crypto, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. A valid hash, signature, or inclusion proof establishes a bounded cryptographic relation, not meaning, authority, safety, or jurisdiction by itself. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] Anders Rundgren, Bret Jordan, and Samuel Erdtman. *JSON Canonicalization Scheme (JCS)*. RFC 8785, 2020. https://www.rfc-editor.org/rfc/rfc8785 **Relationship:** Adjacent canonicalization standard relevant to stable cryptographic commitments.

[**E7**] Ben Laurie et al. *Certificate Transparency Version 2.0*. RFC 9162, 2021. https://www.rfc-editor.org/rfc/rfc9162 **Relationship:** Adjacent append-only Merkle-log standard for inclusion and consistency evidence.

Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Meaning Base (GMB)*. Ableman Corpus

current edition; work identity
AMCS-LOCAL-WORK-D95CCC4FB7B58BFB;
source SHA-256
A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity PAPER.PUB.20257244;
source SHA-256
B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-358AC09F5A13587E;
source SHA-256
AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-091E34BA93992CE4;
source SHA-256
DEB7626CE1AB54F4B541347DCC3C4E14DC4F682230ACCCE1A93D1D536A50C44C.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *The Three Hash Rule.* Ableman Corpus current edition; work identity PAPER.PUB.20788617;
source SHA-256
F6FA73D083B70711A8C817F7779751D1506FEBA0E356DB9805BB1570663731B5.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutional Semantic Hashing: Governed Commitment Schemes for Semantic Provenance, Admissibility, and Constitutional Continuity.* Ableman Corpus current edition; work identity PAPER.PUB.20248734;
source SHA-256
B1DAC3E2CD4D57BAFBF6462BA6ACFFF7CF120CDB7FA37D0C173525744CC0A2F0.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *Semantic Debt.* Zenodo, 2026. doi:10.5281/zenodo.21111666

Adam Ableman *Semantic Manifold Topology.* Zenodo, 2026. doi:10.5281/zenodo.20257244

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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